

Rail Engineering Mathematics

Tutorial Sheet 3 Solutions

1. A cylinder is filled with water to a height of 50 cm. The cylinder has a hole at the bottom which is covered with a stopper. The stopper is released at time $t = 0$ seconds and is allowed to empty. The following data shows the height of the water in the cylinder at different times:

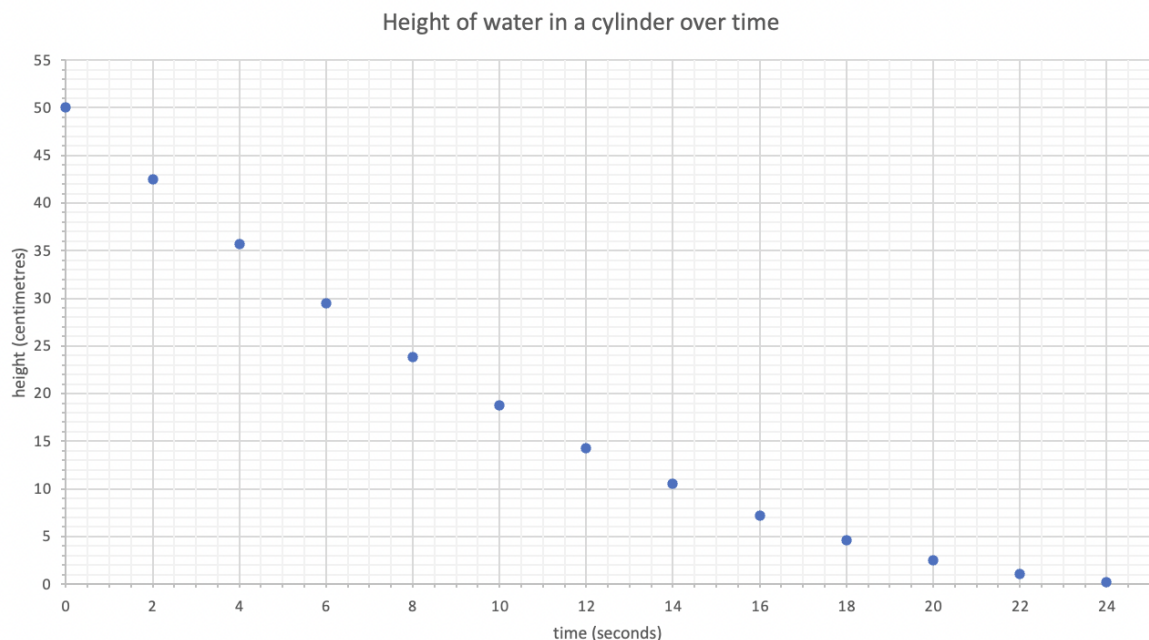
Time (seconds)	Height (cm)
0	50
2	42.5
4	35.7
6	29.5
8	23.8
10	18.8
12	14.3
14	10.5
16	7.2
18	4.6
20	2.5
22	1.1
24	0.2

Depicted in the graph below is the same data:

Assume that the relationship between the height of the water and the time is linear.

(a) With a ruler, draw a linear line of best fit on the graph and determine the equation of the line.

(b) Using the equation obtained in part (a) calculate the height of the water at $t = 1$ s, $t = 12$ s and $t = 24$ s.



(c) What can you conclude about the linear fit in comparison to the original data?

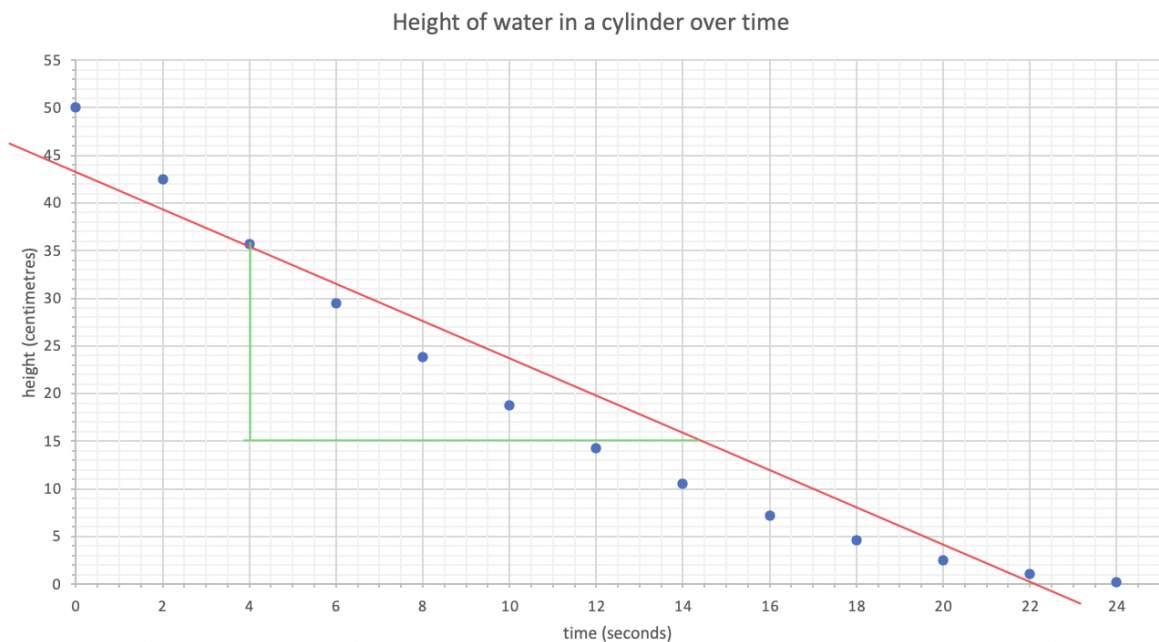
(d) Suggest a better model for the data, determine the equation by hand and test its viability plot plotting the equation and original data on the same axes (This data is available on Blackboard in the file “Tutorial4DataQ1.xlsx”).

Solution:

(a) Note that there are an infinite number of ways that the line of best fit could be drawn and therefore an infinite number of equations could be calculated.

From this line of best fit, we are able to determine the equation of the straight line $h = mt + c$, where h is the height (in cm) and t is the time (in seconds).

The h intercept can be taken directly from the graph: $c \approx 43$. The gradient can be calculated as follows:



$$\begin{aligned}
 m &= \frac{\Delta h}{\Delta t} \\
 &= \frac{15 - 36}{14.5 - 4} \\
 &= \frac{-21}{10.5} \\
 m &= -2
 \end{aligned}$$

Therefore the equation for the line of best fit is $h = -2t + 43$.

(b) The height of the water at $t = 1$ is $h = -2 \times 1 + 43 = 41$ cm.

The height of the water at $t = 12$ is $h = -2 \times 12 + 43 = 19$ cm.

The height of the water at $t = 24$ is $h = -2 \times 24 + 43 = -5$ cm.

(c) Fitting a linear line to the data is not ideal, as the data forms a curve. From the graph alone, it can be seen that for $t \leq 4$ the model heights are lower than the actual heights recorded. for $6 \leq t \leq 20$ the model provides an over-estimate

of the height of the water. For $t > 22$ the model provides a negative value for the height of the water, which is unrealistic.

(d) The original data points are curved which suggests that an exponential model or a quadratic model would be a better model. To check which of these models would be most accurate we can look at the patterns between the original data points.

A linear model: Here equal increments in x yield equal increments in y (arithmetic sequence). We have a constant increment of 2 in the t data the difference between the values is inconsistent. This is why a linear model is not a good fit.

An exponential model: Here there is a common ratio between each y value (given equal increments in x) (geometric sequence). From the table, the first ratio is $\frac{42.5}{50} = 0.85$, the second is $\frac{35.7}{42.5} = 0.84$, the third is $\frac{29.5}{35.7} = 0.83$, etc.

A quadratic model: Data from quadratic functions have a different pattern from arithmetic and geometric sequences. If the data has a common second difference, then it can be modelling by a quadratic function.

Time (seconds)	Height (cm)	1st difference	2nd difference	common ratio
0	50			
2	42.5	-7.5		0.85
4	35.7	-6.8	0.7	0.84
6	29.5	-6.2	0.6	0.83
8	23.8	-5.7	0.5	0.81
10	18.8	-5	0.7	0.79
12	14.3	-4.5	0.5	0.76
14	10.5	-3.8	0.7	0.73
16	7.2	-3.3	0.5	0.69
18	4.6	-2.6	0.7	0.64
20	2.5	-2.1	0.5	0.54
22	1.1	-1.4	0.7	0.44
24	0.2	-0.9	0.5	0.18

There is a fairly consistent 3rd difference, which would indicate that a cubic model would be a good fit. This, however, is beyond the scope of this module.

The differences and ratios do not indicate a clear model, but the quadratic model may yield the better results.

Quadratic model - A quadratic is in the form $h = at^2 + bt + c$. From the graph we know that $c = 50$. Therefore the equation becomes $h = at^2 + bt + 50$. We now have one equation and two unknowns, so we must use two coordinates from the original data.

Substituting (2, 42.5) and (22, 1.1) gives the two equations:

$$42.5 = 4a + 2b + 50 \quad (1) \quad \text{and} \quad 1.1 = 484a + 22b + 50 \quad (2)$$

subtracting 50 from both sides gives:

$$-7.5 = 4a + 2b \quad (1) \quad \text{and} \quad -48.9 = 484a + 22b \quad (2)$$

Multiplying (1) by 11 gives

$$-82.5 = 44a + 22b \quad (3) \quad \text{and} \quad -48.9 = 484a + 22b \quad (2)$$

Subtracting equation (2) from equation (3) gives

$$-33.6 = -440a$$

$$a = \frac{-33.6}{-440}$$

$$a = 0.076$$

Substituting this value of a into equation (1) gives:

$$-7.5 = 4 \times 0.076 + 2b$$

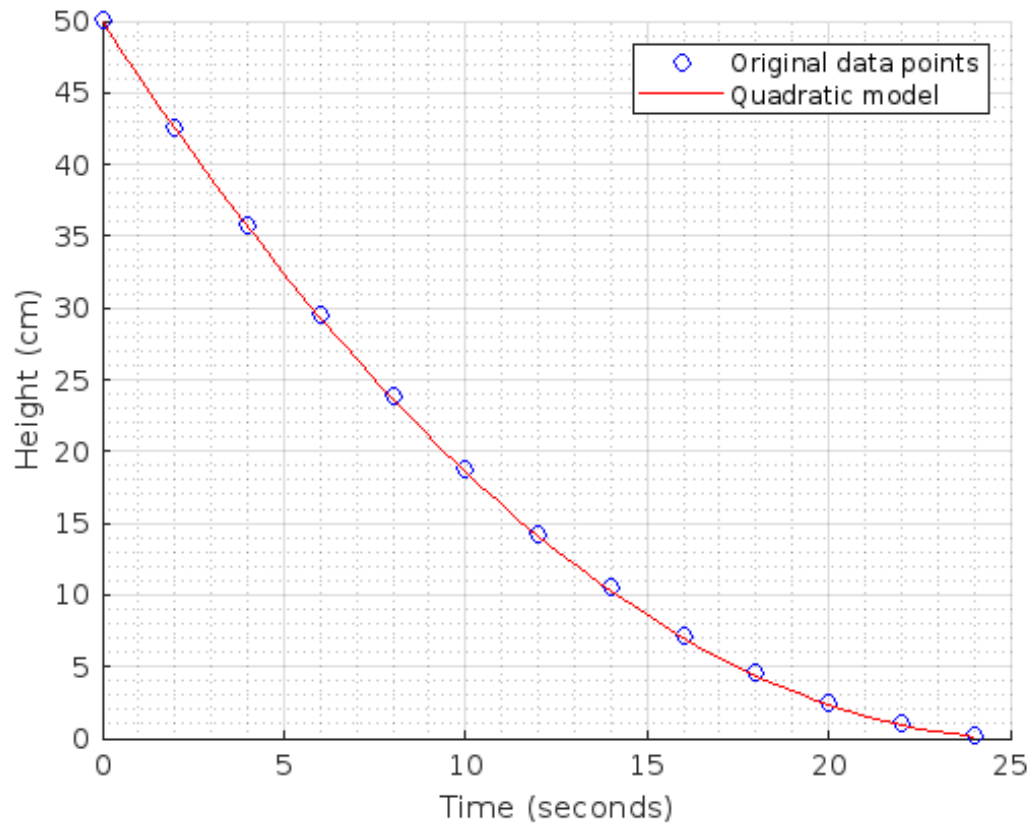
$$-7.5 = 0.304 + 2b$$

$$-7.804 = 2b$$

$$b = \frac{-7.804}{2}$$

$$b = -3.902$$

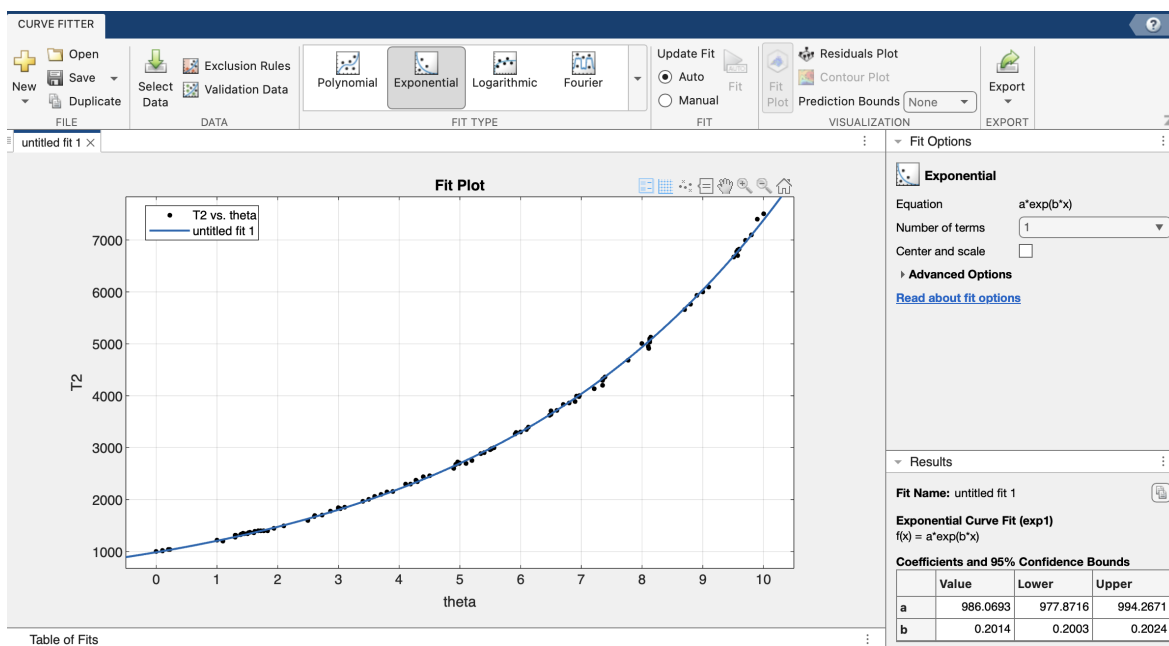
Therefore the quadratic equation representing the curve in the track is $h = 0.076t^2 - 3.902t + 50$. Shown in the graph below is the original data and the quadratic model curve. We can see that the model fits the data well.



2. A pulley is driven by a flat belt with an angle of lap $\theta = \frac{2\pi}{3}$. T_2 is the tension, in N, at one end of the belt. The tension at the other end of the belt remains fixed. Data has been collected, via a series of experiments, by increasing the angle of lap and measuring the tension.

Using MATLAB's `cftool` feature, determine a model that best describes the data (This data is available on Blackboard in the file "Tutorial3DataQ2.xlsx").

Solution:



A good model could be $\theta = 986.07 e^{0.2014 T_2}$.

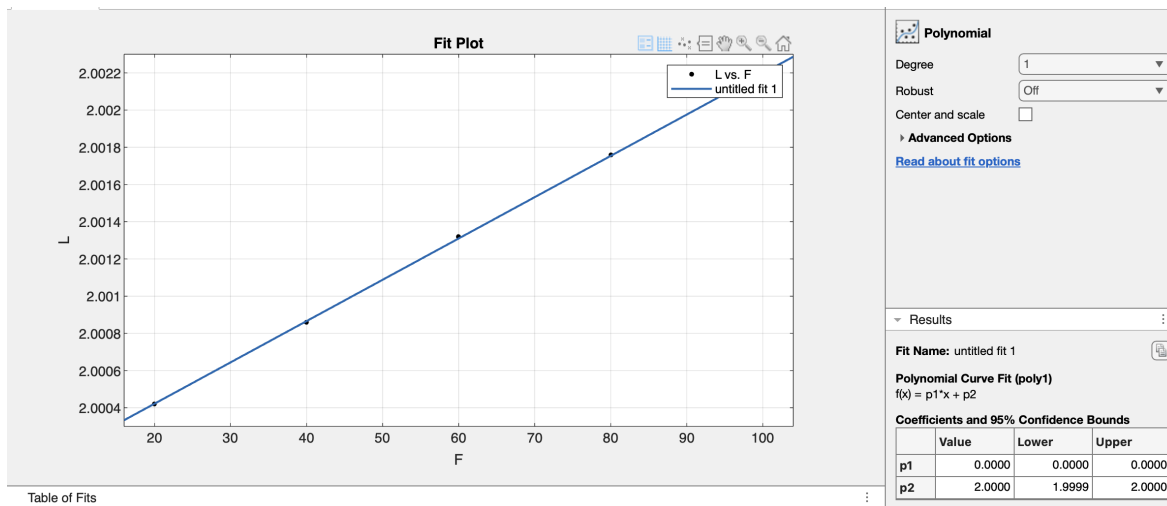
Use MATLAB's `cftool` to fit curves to the data sets below, then state the relationship you were able to establish between the variables. You may need to refer to the “Interactive Functions Explorer” on the module Blackboard site when trying to decide on a suitable function for each problem.

Note: some of the data permit numerous relationships to be explored and established. All original data can be found in individual spreadsheet files.

3. An experiment on the length L of a steel wire under various loads F yields the following data:

F (N)	L (m)
20	2.00042
40	2.00086
60	2.00132
80	2.00176
100	2.00219

Solution:

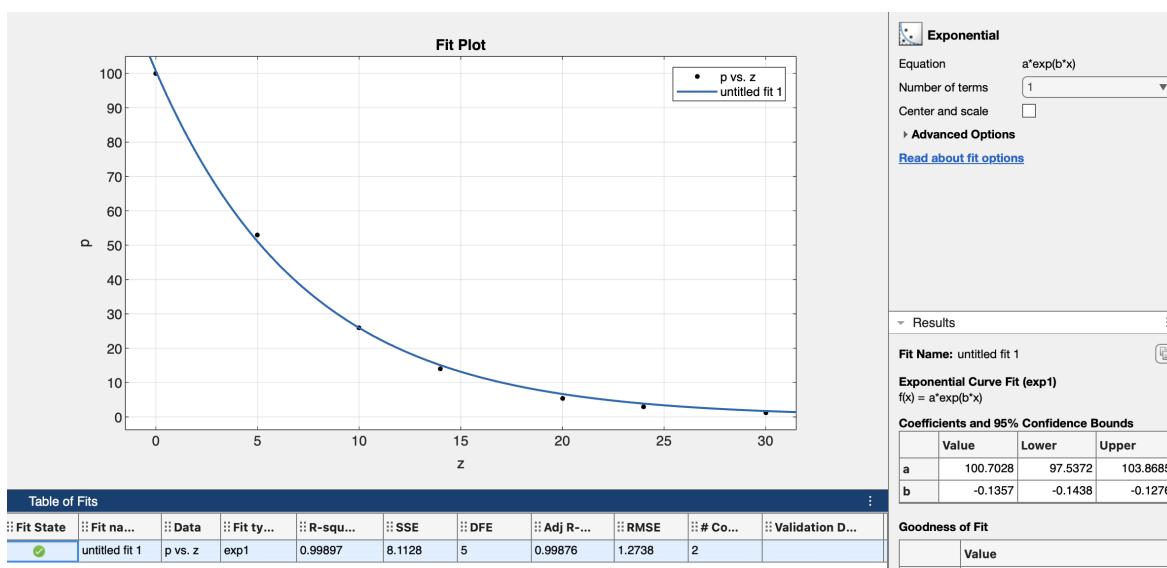


A good model could be $L = 2.22 \times 10^{-5} F + 2$.

4. Physical tables give the variation of atmospheric pressure p with changing altitude z as follows (p_0 represents sea-level pressure):

z (km)	p (% of p_0)
0	100
5	53
10	26
14	14
20	5.4
24	2.9
30	1.2

Solution:

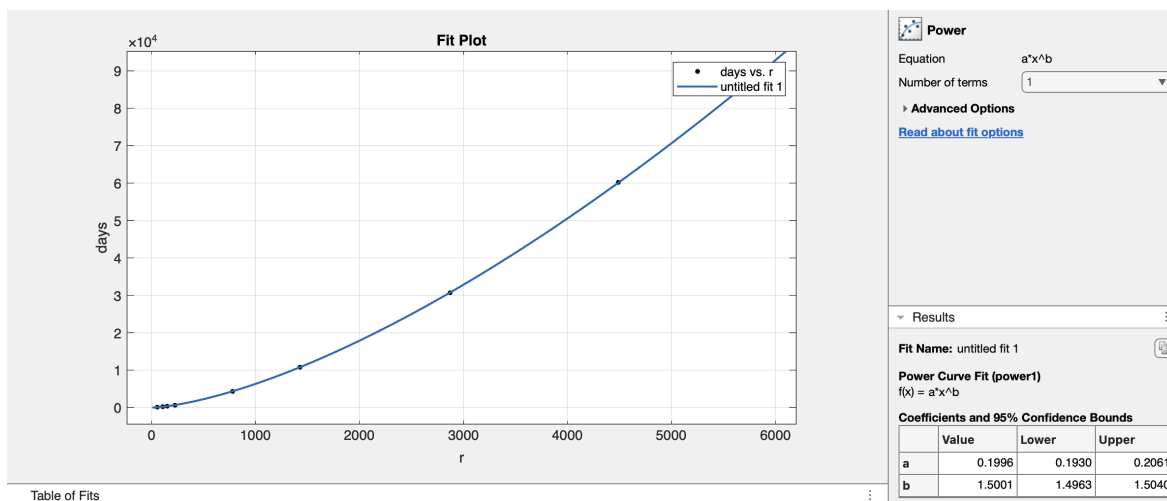


A good model could be $p = 100.71 e^{-0.136z}$.

5. Johannes Kepler spent much time measuring the rotation period T of the planets about the Sun and their mean distances r from the Sun:

Planet	r (Gm)	T (days)
Mercury	57.9	88
Venus	108.2	225
Earth	149.6	365
Mars	227.9	687
Jupiter	778.3	4329
Saturn	1427	10753
Uranus	2870	30660
Neptune	4487	60150
Pluto	5907	90670

Solution:



A good model could be $T = 0.2r^{1.5}$.

6. An experiment has been carried out to determine the relationship between the mean flow velocity V (m/s) and the pressure drop per unit length ΔP (Pa/m) across a short section of a smooth circular pipe. For laminar flow in a constant-diameter pipe the pressure drop is expected to be directly proportional to the mean velocity. The experiment validates a pipeline sensor design for low-speed flows.

A steady airflow is driven through the test pipe in a wind-tunnel rig. At each set mean velocity V , the pressure drop per unit length ΔP is measured using a calibrated differential pressure transducer. To account for small flow disturbances and sensor noise, four repeated measurements are taken at each velocity. All data are recorded in the spreadsheet “Tutorial3DataQ6.xlsx”.

(a) Plot all data points on a single scatter graph with the pressure drop per unit length ΔP on the vertical axis and the mean velocity V on the horizontal axis. Discuss the physical meaning of the trend shown in the graph.

Solution:

The graph is in MATLAB.

In terms of the observed trend, the points lie very close to a straight line passing through the origin region — i.e. ΔP increases as V increases. At this stage the functional relationship is not yet established; the scatter plot only shows an empirical trend. We must fit models and quantify goodness-of-fit to determine which mathematical form best describes the data.

In terms of the physical context and expectations: For laminar flow in a circular pipe theory (Poiseuille-like behaviour) predicts ΔP per unit length is directly proportional to V (a linear dependence), whereas turbulent flow usually gives a stronger, nonlinear dependence (roughly $\propto V^2$ in many engineering correlations). The observed straight-line trend therefore suggests laminar-like behaviour, but we must confirm by model fitting and by checking whether the flow regime assumptions (Reynolds number, pipe geometry) hold.

(b) Using MATLAB’s Curve Fitting Tool (cftool), fit the following mathematical models to all data points:

- A Linear model
- A Quadratic model
- A Power model

Report the equation and the R^2 value for each of the three models.

Solution:

Select “stacked_velocity” for the X data and “DelP_all” for the Y data.

Using all data points, three models were fitted:

Linear model: $\Delta P = 150.2714V - 0.02$ (Pa/m) with $R^2 = 0.99991$.

Quadratic model: $\Delta P = 1.875V^2 + 148.9589V + 0.155$ with $R^2 = 0.99992$.

Power model: $\Delta P = 150.358V^{1.0011}$ with $R^2 = 0.99992$.

All three fits produce very high R^2 values (≈ 0.9999) because the data follow an almost perfect linear relationship ($\Delta P \approx 150V$) with only small measurement scatter. The power-fit exponent is effectively 1, making it numerically equivalent to the linear model. The quadratic fit adds a very small quadratic term, which improves R^2 insignificantly and lacks physical justification for laminar flow.

(c) Plot all raw data points and the three fitted models from part (b) on a single graph, over the velocity range $V = 0$ to $V = 1.2$ m/s.

Based on both the statistical indicator (R^2) and the physical implications of each fitted curve up to 1.2 m/s, decide which mathematical model best represents the experimental data. Discuss the strengths and weaknesses of your chosen model and critically evaluate limitations of using it for extrapolation beyond the measured range.

Solution:

All three models give nearly identical R^2 values, so statistical indicators alone cannot distinguish between them. The selection must therefore rely on physical reasoning.

- The **linear model** is the physically expected relationship for laminar flow in a constant-diameter pipe, as predicted by Poiseuille-type flow theory where $\Delta P \propto V$.
- The power model essentially reduces to the same form because $b \approx 1$.

- The quadratic model fits numerically but introduces an unnecessary term without physical basis.

The linear model best represents the data because it aligns with both the empirical results and the theoretical relationship for laminar flow.

Strengths and weaknesses of the chosen model:

Strengths:

- Physically grounded: Consistent with the theoretical expectation for laminar flow ($\Delta P \propto V$).
- Parsimonious: Simple two-parameter form (m, c) with minimal risk of overfitting.
- Interpretable: The slope has a clear physical meaning related to viscosity and pipe geometry.

Weaknesses:

- Valid only under laminar-flow conditions; will fail if the flow becomes transitional or turbulent.
- Small non-zero intercept arises from experimental noise, not a real physical offset.
- Systematic errors (e.g. temperature effects or sensor bias) could slightly distort the fitted slope.

Limitations and extrapolation caution:

- **Flow regime change:** The linear relationship holds only while flow remains laminar. If the Reynolds number exceeds the laminar limit ($Re \approx 2300$), the pressure–velocity dependence becomes nonlinear.
- **Geometry and fluid changes:** Extrapolation to other pipe diameters or fluids invalidates the constant of proportionality.
- **Instrumentation limits:** Sensor calibration and measurement range may restrict valid extrapolation.
- **Mathematical behaviour:** Quadratic or power models may diverge outside the fitted range, reinforcing the need to choose the physically justified linear model for safe extrapolation.

In summary the data strongly support a linear relationship within the measured range. The linear model is both statistically excellent and physically justified; extrapolation should only be done within the laminar-flow regime.